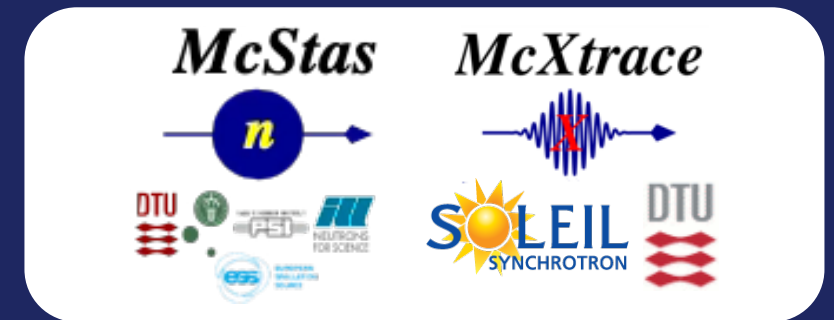




Challenge

Team challenge:

From McStas/McXtrace Data to AI



PaNRAID School · supported by PEPR DIADEM

Challenge of the week



You will work in groups of 4. We have mixed you to initiate some networking ;)

Tasks throughout the week:

- Talk to your group members and **define** a problem of interest from an AI point of view.
- Use an **instrument** in McStas/McXtrace to tackle the previously defined problem.
- **Generate** a synthetic dataset varying instrument and sample parameters.
- **Train** an AI model to tackle the problem of interest.
- **Present** your results! < 20 min presentation on Friday

Timeline



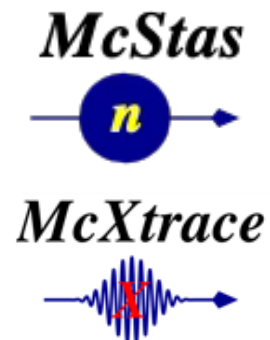
Monday

Tuesday

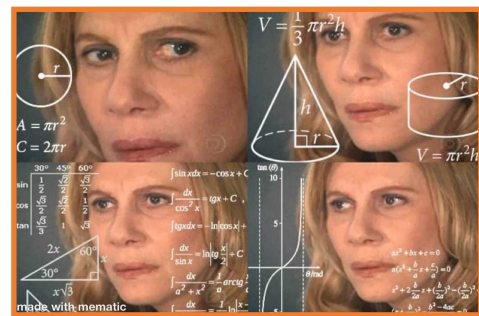
Wednesday

Thursday

Friday



SAMPLES.



Challenge of the week



On the problem

- Regression, classification, inverse, anomaly detection, unsupervised learning, etc.

On the dataset generation

- Generate (labeled or not) data by varying meaningful parameters.

On the AI model and training

- Split the data in train and test partitions and use them accordingly.
- Keep it simple to have something working. Then complicate it as much as you want.

On the results

- Explain what worked, what failed, and what the results mean.
- Think on future perspectives.

Scope that fits the time



Aim for a convincing minimum result before adding complexity

Good scope

- One target variable or a small class set
- A controlled parameter sweep
- One or two architectures to compare and analyze results
- A reproducible train / validation / test split

Risky scope

- A full instrument digital twin
- Millions of expensive simulations
- Many architectures with little analysis

***Target a first end-to-end result by Thursday night
polish results and prepare slides on Friday***

Project directions



Use these as starting points, not fixed assignments

DIRECTION	SCIENTIFIC QUESTION	AI FORMULATION
Fast surrogate	Predict intensity or a spectrum from instrument parameters.	Regression
Inverse problem	Infer sample or instrument parameters from simulated detector output.	Regression / uncertainty
State recognition	Identify sample phase, structure, geometry, or instrument configuration.	Classification
Fault or anomaly	Detect misalignment or unusual detector patterns.	Anomaly detection

We are always available for questions

McStas



McXtrace



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